1 BIC ATAC-seq Pipeline (v2.1.0)

1.1 Introduction

The results are delivered in one folder, `atacSeq`. `macs` holds the peaks called for each sample. `atlas` holds the merged peak atlas of the project and the raw count matrix over it, which is the input to any normalization or differential analysis. `bigwig` holds a normalized signal track per sample for a genome browser. `metrics` holds the QC metrics and plots. Projects with replicates have a `diff` folder with the differential accessibility analysis and, when requested, its pathway enrichment (GSEA) and the enrichment of transcription factor motifs in its regions. `docs` holds this document (`output.pdf`, `output.html`) and the methods document (`methods.pdf`, `methods.html`), which describes how each file was made.

In the names below, `<sample>` is the sample ID of the BAM and `<no>` is the project number (for example `18143`). Coordinates are 0-based with an exclusive end (BED convention) unless a file says otherwise. Peaks, the atlas and the signal tracks cover the autosomes and the X and Y chromosomes only; mitochondrial, unplaced and other contigs are left out.

1.2 Directory Structure

```
atacSeq
├── atlas
│   ├── macsPeaksMerged.saf
│   ├── peaks_raw_fcCounts.txt
│   ├── peaks_raw_fcCounts.txt.summary
│   ├── peaks_raw_fcCounts.checks.txt
│   └── scaleFactorsDESeq2.csv
├── bigwig
│   └── <sample>_postProcess.shifted.10mNorm.bw
├── macs
│   └── <sample>_postProcess.shifted
│       ├── <sample>_postProcess.shifted_peaks.narrowPeak
│       ├── <sample>_postProcess.shifted_summits.bed
│       ├── <sample>_postProcess.shifted_peaks.xls
│       ├── <sample>_postProcess.shifted_peaks.blacklistFiltered.narrowPeak
│       └── <sample>_postProcess.shifted_summits.blacklistFiltered.bed
├── metrics
│   ├── Proj_<no>__postInsDistribution.pdf
│   ├── Proj_<no>__ATACSeqQC.pdf
│   ├── Proj_<no>__ATACSeqDepth.csv
│   ├── <sample>_postProcess___INS.txt
│   ├── <sample>_postProcess___INS.pdf
│   ├── <sample>_postProcess___MITO.txt
│   ├── <sample>_postProcess.tss_enrich.csv
│   ├── <sample>_postProcess.tss_enrich.pdf
│   └── <sample>_postProcess.large_tss_enrich.png
```

```
├── docs
│   ├── output.pdf
│   ├── output.html
│   ├── methods.pdf
│   └── methods.html
└── diff                        (projects with replicates only)
    ├── Proj_<no>__DiffPeaksEdgeR_V5.xlsx
    ├── Proj_<no>__DiffPeaks_V5.pdf
    ├── Proj_<no>__PathwayGSEA.xlsx    (when requested)
    ├── Proj_<no>__PathwayGSEA.pdf     (when requested)
    ├── Proj_<no>__PathwayGSEA_Dot.pdf         (when requested)
    ├── Proj_<no>__PathwayGSEA_Heatmap.pdf     (when requested, 2+ comparisons)
    ├── Proj_<no>__PathwayGSEA_Enrichment.pdf  (when requested)
    ├── Proj_<no>__MotifEnrichment.xlsx        (when requested)
    └── Proj_<no>__MotifEnrichment.pdf         (when requested)
```

The filtered BAM files (`<sample>_postProcess.bam`) are not delivered. If you need them, please contact us.

1.3 Peaks (`macs`)

One folder per sample with the output of MACS2 and two files derived from it. The folder and file names carry the suffix `_postProcess.shifted` because the peaks were called on the filtered, Tn5-shifted reads.

MACS2 output. These three files are MACS2's own, delivered unchanged:

- `*_peaks.narrowPeak`: every peak MACS2 called at $p < 0.01$, in ENCODE narrowPeak format (BED6+4). The columns are chromosome, start, end, peak name, score, strand (always `.`), fold enrichment at the summit, -log10 p-value, -log10 q-value, and the position of the summit relative to the peak start. A peak with several summits appears once per summit, with a letter suffix on its name (`_peak_12a`, `_peak_12b`).
- `*_summits.bed`: one line per summit, the single base of highest signal in each peak, with the peak name and its -log10 p-value.
- `*_peaks.xls`: the same peaks as a tab-delimited text table (not an Excel file) with 1-based coordinates, peak length, summit position, pileup, -log10 p-value, fold enrichment and -log10 q-value. The header lines record the MACS2 command and settings.

Filtered peaks. The same peaks after the ENCODE post-processing steps (see `methods.pdf`): at most 300,000 peaks per sample, the most significant by p-value, clipped to the chromosome ends, with every peak that overlaps the ENCODE blacklist removed and the score capped at 1000. Peaks keep their MACS2 names, so a peak can be looked up in the three files above.

- `*_peaks.blacklistFiltered.narrowPeak`: the filtered peaks, same columns as `*_peaks.narrowPeak`, sorted by p-value.
- `*_summits.blacklistFiltered.bed`: the summits of the filtered peaks.

Use the `blacklistFiltered` files for any per-sample peak analysis. They are the peaks the atlas was built from. The per-sample peak counts in these files are the ones to read against ENCODE's data

standards.

1.4 Peak Atlas and Counts (`atlas`)

1.4.1 Atlas (`macsPeaksMerged.saf`)

The peak atlas: the filtered peaks of all samples pooled and merged, with peaks within 500 bp of each other joined into one region. The atlas is one set of regions shared by every sample, so that the samples can be compared region by region. It is in SAF format, as read by featureCounts, with no header line and five tab-separated columns:

Column	Contents
`GeneID`	region name, `Peak_1`, `Peak_2`, … in genome order
`Chr`	chromosome
`Start`	start, 1-based
`End`	end, inclusive
`Strand`	always `+`

Note that SAF coordinates are 1-based and inclusive, unlike the BED and narrowPeak files. To convert a region to BED, subtract 1 from `Start`.

Because the atlas pools and merges the peaks of all samples, its number of regions grows with the number of samples and is not a per-sample peak count.

1.4.2 Raw counts (`peaks_raw_fcCounts.txt`)

The number of fragments of each sample in each atlas region, from featureCounts. The first line is a comment with the featureCounts command; the second is the header. The columns are the five atlas columns (`Geneid`, `Chr`, `Start`, `End`, `Strand`), the region `Length`, and then one column per sample, named by the path of its filtered BAM (`out/<sample>/<sample>_postProcess.bam`).

A fragment is a read pair; it is counted once in every region it overlaps by at least one base, so a fragment that spans two adjacent regions is counted in both. The counts are raw, not normalized. They are the input to the DESeq2 scale factors, the QC plots and the differential analysis, and are what to use for any analysis that needs counts.

1.4.3 Count summary (`peaks_raw_fcCounts.txt.summary`)

For each sample, the number of fragments assigned to the atlas (`Assigned`) and the number not assigned, by reason (the `Unassigned_*` rows; `Unassigned_NoFeatures` is fragments that overlap no atlas region). The sum of the column is the number of fragments in the filtered BAM.

1.4.4 Count checks (`peaks_raw_fcCounts.checks.txt`)

The two checks the counts passed before they were used (see `methods.pdf`), two lines per sample:

- `CHECK_A`: the sample's fragment total in the count summary (`frag`) against half its mapped reads (`pairs`). The `ratio` must be between 0.95 and 1.05.
- `CHECK_B`: the fragments featureCounts assigned to the atlas (`assigned`) against the same number counted with samtools (`samtools`). The difference (`diff`) must be 0.

The last line is the overall result. A run is delivered only if every line is `OK`.

1.4.5 Scale factors (`scaleFactorsDESeq2.csv`)

One row per sample, named by the path of its filtered BAM as in the count matrix (with `/` and `-` replaced by `.`), and one column, `scaleFactor`: the reciprocal of the DESeq2 size factor computed from the raw counts. A normalized count is the raw count multiplied by the sample's scale factor. No delivered file has them applied; the signal tracks are scaled to 10 million reads instead.

1.5 Signal Tracks (`bigwig`)

`<sample>_postProcess.shifted.10mNorm.bw` is the coverage of the filtered, Tn5-shifted reads of the sample, scaled to 10 million reads (`10mNorm`), in bigWig format. The tracks can be loaded directly into IGV or the UCSC Genome Browser. Because every sample is scaled to the same number of reads, the tracks can be compared by eye across samples. The value at a base is the number of shifted reads covering it, multiplied by 10 million and divided by the sample's total number of shifted reads.

1.6 Metrics (`metrics`)

1.6.1 QC report (`Proj_<no>__ATACSeqQC.pdf`)

Six pages built from the count matrix, with samples colored by group:

1. The percentage of each sample's fragments that fall in the atlas (`InPeaks`) against those outside it. This is a signal-to-noise measure; a sample with a much lower percentage than the others may have failed. It is computed against the pooled atlas, and its denominator includes mitochondrial fragments, so it is not ENCODE's per-sample FRiP and should not be read against ENCODE's FRiP standard.
2. The same as counts, in millions of fragments.
3. Sequencing depth in peaks: each sample's fragments in the atlas, its ratio to the median sample, and the share of the peaks used for the PCA in which it has fewer than 5 fragments. A sample below half the median is marked `LOW DEPTH`. Such a sample has noisy counts and may separate from all others on a principal component of its own, whatever its biology.
4. A principal component analysis of the samples on the normalized log2 counts over the atlas, with sample labels. The size of a point is the sample's fragments in peaks. Replicates of a

group should lie together; samples that separate from their group on PC1 or PC2 are worth a look.
5. The same PCA without labels.
6. A depth check of the PCA: beside the PCA of page 4, the same PCA after every sample is reduced, by drawing its fragments at random, to about the depth of the smallest. A sample that is apart from its group on the left but not on the right is apart because of its depth; a sample apart on both differs from its group in its data. The reduced counts are used for this page only.

1.6.2 Depth table (`Proj_<no>__ATACSeqDepth.csv`)

The numbers of page 3, one row per sample: `SampleID`, `Group`, `FragmentsInPeaks`, `RatioToMedian`, `FracKeptPeaksLT5` (the share of the peaks used for the PCA with fewer than 5 fragments) and `LowDepth` (`TRUE` below half the median).

1.6.3 Insert-size distributions (`Proj_<no>__postInsDistribution.pdf`)

The insert-size (fragment-length) distribution of each sample's filtered reads, from Picard CollectInsertSizeMetrics:

1. All samples on one plot, density against insert size (0 to 1000 bp), colored by group with one line type per replicate.
2. The same on a log scale, which shows the nucleosomal peaks.
3. One panel per sample, six per page.

A successful ATAC-seq library shows a large nucleosome-free peak below about 100 bp and smaller peaks at about 200 bp (one nucleosome) and 400 bp (two nucleosomes).

The histogram covers 0 to 1000 bp. Its smallest insert is 31 bp: pairs with an insert of 30 bp or less are removed by the read filter (see the methods document).

The per-sample files behind the plot are `<sample>_postProcess___INS.txt`, the Picard metrics file (summary statistics such as the median and mode insert size, followed by the histogram as `insert_size` and count columns), and `<sample>_postProcess___INS.pdf`, Picard's own histogram plot.

1.6.4 Mitochondrial fraction (`<sample>_postProcess___MITO.txt`)

A header line and one line for the sample, with the columns `SAMPLE`, `MITO_CONTIG`, `READS` (all reads of the filtered BAM), `MITO_READS` (those on the mitochondrial contig) and `FRAC_MITO`. Mitochondrial reads are left out of the peaks, the atlas and the signal tracks. The fraction is computed after the read filter, which removes duplicates, so it is lower than a fraction computed on the raw alignments (as ENCODE reports it) and should not be read against that standard.

1.6.5 TSS enrichment (`<sample>_postProcess.tss_enrich.csv`)

The ENCODE TSS enrichment score, the standard measure of signal-to-noise for ATAC-seq: how much higher the signal is at transcription start sites than in the flanking regions 2 kb away. Columns:

Column	Contents
`sample_id`	sample
`tss_enrich`	the TSS enrichment score
`peak_pos_bp`	position of the maximum of the profile, relative to the TSS
`avg_noise`	mean signal in the flanking regions, the denominator of the score
`n_tss_used`, `n_tss_total`, `pct_tss_used`	TSSs scored, TSSs in the reference set, and their ratio; `pct_tss_used` should be close to 1
`n_mapped_reads`	reads of the filtered BAM
`read_len`	read length used for the computation
`bam_file`	the filtered BAM

ENCODE's reading of the score depends on the genome and on the TSS annotation. The scores here use ENCODE's GENCODE TSS sets (v24 for hg38, v19 for b37, vM7 for mm10); read the cutoffs as a guide:

Genome	Concerning	Acceptable	Ideal
hg38	< 5	5 to 7	> 7
b37	< 6	6 to 10	> 10
mm10	< 10	10 to 15	> 15

Two plots accompany the score: `<sample>_postProcess.tss_enrich.pdf`, the mean signal profile across the 4 kb window centered on the TSS, and `<sample>_postProcess.large_tss_enrich.png`, a heatmap of the signal at every TSS, sorted by signal, with the mean profile above it.

1.7 Differential Accessibility

For projects with replicates, pairwise comparisons between groups are delivered in the `diff` folder as two files, named with the project number and, when one was used, a run tag:

- `Proj_<no>__DiffPeaksEdgeR_V5.xlsx`: one `Summary` sheet with the number of significant regions per comparison, then one sheet per comparison, named `<group2>_v_<group1>`, listing the regions significant at FDR < 0.05. Columns: `PeakNo` (the atlas region), `logFC` (log2 fold change, group 2 over group 1: positive means more accessible in group 2), `logCPM` (mean log2 counts per million), `PValue`, `FDR` (Benjamini-Hochberg adjusted p-value), the region's `Chr`, `Start`, `End` (1-based, from the atlas), `Strand` and `Length`, and the ChIPseeker annotation: `SYMBOL` and `GENENAME` of the nearest gene, `annotation` (promoter, exon, intron, distal intergenic, …) and `distanceToTSS`.
- `Proj_<no>__DiffPeaks_V5.pdf`: the PCA of the samples (with and without labels; the point size is the sample's fragments in peaks, as in the QC report), then for each comparison an MA plot (log fold change against mean log CPM) and a volcano plot (log fold change against p-value, on a log scale). Regions significant at FDR < 0.05 are colored by the group they are

more accessible in, with the count of each in the legend; the dashed line on the volcano plot is the p-value at which the FDR cut falls.

The names above are those without a run tag (hence the double underscore); with a tag `<tag>` they are `Proj_<no>_<tag>_DiffPeaksEdgeR_V5.xlsx` and `Proj_<no>_<tag>_DiffPeaks_V5.pdf`.

Samples whose group starts with `EXC` were excluded from the differential analysis.

In our deliveries made before October 2026 the `FDR` column was a Holm correction, which is much stricter; the same data now give more regions at FDR < 0.05.

1.7.1 Pathway enrichment (`Proj_<no>__PathwayGSEA.xlsx`)

When requested, each comparison is also tested for pathway enrichment (GSEA) against MSigDB gene sets. Every tested region is assigned to each gene with a TSS within a set distance of it (by default 1 kb, i.e. the genes whose promoter it is in; a region at a promoter shared by two genes counts for both), and each gene is scored by its region with the strongest change: sign(log fold change) x -log10(p-value). The `Parameters` sheet gives the distance and the collections used. The genes ranked by this score are tested against each collection:

- `Summary`: per comparison and collection, `GenesRanked`, `SetsTested` (sets of 15 to 500 ranked genes), and the number of sets with adjusted p < 0.05 and a positive (`SigPositiveNES`) or negative (`SigNegativeNES`) enrichment score, and how many of those are independent (`SigIndependent`, below).
- One sheet per comparison, named as in the differential analysis, with every gene set tested, sorted by p-value. **These sheets are NOT filtered for adjusted p < 0.05**: most rows are not significant. Filter on `padj` to find the significant sets (the differential analysis sheets, by contrast, list only significant regions). Columns: `Collection` (e.g. `H`, `C2:CP`, `C5:GO:BP`), `Subcollection` (e.g. `CP:REACTOME`), `pathway` (the MSigDB set name), `NES` (normalized enrichment score: positive means the set's genes are more accessible in group 2, negative in group 1), `pval`, `padj` (Benjamini-Hochberg within the collection), `log2err` (the error of the p-value estimate), `ES` (enrichment score), `size` (ranked genes in the set), `CollapsedInto` (below) and `leadingEdge` (the genes that drive the enrichment). A few sets may have no `NES`, `pval` or `padj`: the p-value could not be estimated for them.
- `Parameters`: the settings and the MSigDB and software versions.

Gene sets overlap, and one strong set makes its overlapping sets significant too. The significant sets of each comparison and collection are therefore reduced to independent ones (fgsea's `collapsePathways`): going from the smallest p-value, a set that is no longer significant once the genes of a more significant set are taken into account is redundant, and `CollapsedInto` names that set. `CollapsedInto` is empty for an independent set and for a set that is not significant. **The plots show only independent sets**; the sheets keep every set.

The plots, each in its own PDF, show only sets with adjusted p < 0.05. A comparison and collection with no such set has no page; the `Summary` sheet gives 0 for it.

- `Proj_<no>__PathwayGSEA.pdf`: one page per comparison and collection, a bar chart of the NES of the independent sets, at most the 15 with the smallest p-value on each side of 0, colored by the group they are more accessible in.

- `Proj_<no>__PathwayGSEA_Dot.pdf`: the same sets as a dot plot; the dot size is the number of genes in the set, its color the adjusted p-value as -log10(padj), from light to dark, with the scale clipped to the 5th-95th percentile of the page.
- `Proj_<no>__PathwayGSEA_Heatmap.pdf` (with two or more comparisons): one page per collection, the NES of each set in every comparison. The rows are the sets independent and significant in at least one comparison (at most 50, those with the smallest adjusted p-value), clustered on their NES; a cell is grey where the set is not significant in that comparison, or was not tested. A positive NES means more accessible in the first group of the comparison's name. Blue is negative and red positive; the color scale ends at the 95th percentile of |NES|, and larger values have the end color.
- `Proj_<no>__PathwayGSEA_Enrichment.pdf`: for each comparison and collection, the running enrichment score of the top independent sets (at most 3 per side of 0), then one page per set with a heatmap of its leading-edge genes: each row is a gene, by the region that gave it its score (at most 50 genes, those with the largest score), each column a sample of the comparison, and the color the region's normalized accessibility (log2 CPM, as tested by edgeR), centered and scaled across the samples (blue below the mean, red above, with the scale ending at the 95th percentile of |z|). Two genes that share a promoter region have the same row values.

Human data use the human MSigDB collections; the default is Hallmark (`H`), the canonical pathways (`C2:CP`: Reactome, WikiPathways, KEGG MEDICUS, BioCarta, PID; not the legacy KEGG sets) and GO biological process (`C5:GO:BP`). Mouse data use the mouse MSigDB (`MH`, `M2:CP`, `M5:GO:BP`). As for the differential analysis, a run tag `<tag>` gives `Proj_<no>_<tag>_PathwayGSEA.xlsx`, and so on.

1.7.2 Motif enrichment (`Proj_<no>__MotifEnrichment.xlsx`)

When requested, the regions of each comparison are also tested for enriched transcription factor binding motifs, the 879 vertebrate profiles of JASPAR 2024 (CORE, non-redundant). Each region is represented by 501 bp centered on its strongest peak summit, the summit with the smallest MACS2 p-value of all the samples' peaks in the region. Two sets of regions are tested per comparison: those significantly more accessible (FDR < 0.05, as in the differential analysis) in group 2 (`up`) and those more accessible in group 1 (`down`). The background of both is the other regions tested in the comparison (FDR $>= 0.05$), weighted to match the set's G+C content and short-sequence (k-mer) composition, so that a motif is not found enriched only because the set is GC-rich, as promoters are. A set of fewer than 50 regions is not tested. The width and the smallest set are defaults; the `Parameters` sheet gives the values used.

- `Summary`: per comparison and direction, `HigherIn` (the group the set is more accessible in), `Peaks` (regions in the set), `Background` (regions in the background), `Status` (`tested`, or why not), `MotifsTested` and the number of motifs enriched with adjusted $p < 0.05$.
- One sheet per comparison with a tested set, named as in the differential analysis, with every motif tested for each set, first `up`, then `down`, each sorted by p-value. **These sheets are NOT filtered for adjusted $p < 0.05$**: filter on `padj` to find the enriched motifs. Columns: `Direction`, `HigherIn`, `MotifID` (the JASPAR ID, e.g. `MA0099.4`; look it up at https://jaspar.elixir.no), `Motif` (the transcription factor, or two for a heterodimer such as `FOS::JUN`), `log2enr` (log2 of the fraction of set regions with a motif hit over that of the weighted background), `pval` (one-sided: more set regions with a hit than expected), `padj` (Benjamini-Hochberg within the

set), `FgWithHit` and `BgWithHit` (percent of set and of weighted background regions with a hit) and `MotifGC` (percent G+C of the motif).
- `Parameters`: the settings and the software versions.

Related transcription factors bind similar motifs, so one change is often seen as a family of enriched motifs: for example AP-1 (FOS, JUN, ATF, BATF and their dimers) usually appear together. The test cannot tell which member of a family is the one binding.

- `Proj_<no>__MotifEnrichment.pdf`: one page per comparison with at least one enriched motif, a heatmap of the log2 enrichment of the top motifs (at most 20 per set, those with the smallest p-value), a column for each set; then a page with the top motifs of all comparisons (at most 50, those with the smallest adjusted p-value), a column for each tested set. The rows are clustered on the enrichment; a cell is grey where the motif has adjusted p >= 0.05 in that set, or the set was not tested. The color scale runs from light to dark and is clipped to the 5th-95th percentile of the page.

A run tag `<tag>` gives `Proj_<no>_<tag>_MotifEnrichment.xlsx` and `.pdf`.